

Worksheet — 1.5 Legal, Moral, Cultural & Ethical Issues

OCR A Level Computer Science H446 — Component 01, Section 1.5 (1.5.1 legislation; 1.5.2 moral/ethical/cultural/social impact)

Name: ____ Date: ____

Time: ~60 min.

Section A — Skills practice

Activity 1 — Match each law to what it covers

Draw a line, or write the matching letter in the box.

Law		What it covers
1. Data Protection Act 2018 (+ GDPR)	☐	A. How public bodies (police, security services) may monitor/intercept communications, with rules on who can authorise it
2. Computer Misuse Act 1990	☐	B. Intellectual property — protects original works, software, music and designs from being copied without a licence
3. Copyright, Designs & Patents Act 1988	☐	C. How organisations collect, store and process personal data about living individuals
4. Regulation of Investigatory Powers Act 2000 (RIPA)	☐	D. Unauthorised access to computer systems — hacking, malware and unauthorised modification

Answers: 1 = _ 2 = 3 = _ 4 =

Activity 2 — Which law applies?

For each scenario, name the relevant **Act**. For any Computer Misuse Act scenario, also state **which of the three offences** applies (Offence 1 = unauthorised access; Offence 2 = unauthorised access *with intent* to commit a further offence; Offence 3 = unauthorised *modification*).

1. A company keeps customers' addresses for ten years after they stop being customers, with no reason to do so. Act: _____
2. A student logs in with a teacher's password "just to look around" the staff system, but changes nothing. Act: ____ Offence (if CMA): _____
3. Someone downloads and shares the latest blockbuster film for free from a torrent site. Act: _____
4. A criminal hacks into a bank's servers in order to transfer money into their own account. Act: ____ Offence (if CMA): _____
5. A local council secretly intercepts a resident's emails without obtaining the proper authorisation. Act: _____

Plan **before** you write. Fill the skeleton with short bullet points. For each point use: *point* → *consequence* → *stakeholder* → *counter-point*. Aim for **3–4+ stakeholders**.

AGAINST the statement (arguments it does more good than harm) - _____ - _____ - _____

Stakeholders I will name: _____

Justified conclusion (must take a side — often with a condition): - _____ - _____

Activity 5 — Short-answer definitions

1. **Personal data:** _____

2. **Data subject:** _____

3. **Copyright** (and how does it differ from a **patent**?): _____

4. **Digital divide:** _____

5. **The difference between GDPR and the DPA 2018:** _____

Activity 6 — Balanced argument builder

Scenario: A national supermarket chain announces it will replace **all staffed checkouts** with AI-powered self-checkouts that use **facial recognition** to identify suspected shoplifters automatically.

Examiners repeatedly report that one-sided answers cap out at the middle band. Train the habit here: for **every** stakeholder you must write **one benefit AND one drawback** — no cell may be left blank. Each entry must be a *developed* point (cause → effect), not a single word.

Stakeholder	Benefit (+) of the change	Drawback (–) of the change
Customers / shoppers		
Checkout staff		
The supermarket (business)		
Government / wider society		

Now the evaluative step — answer in one or two sentences each:

(a) Which stakeholder is affected **most seriously**, and why?

(b) Suggest **one safeguard or condition** under which the change would become more acceptable.

Activity 7 — Precision: exactly which law, exactly which offence?

Vague answers such as "data protection laws" or "the hacking law" score **zero** in the exam. For each scenario write the **full name of the Act (including the year)** and, where the Computer Misuse Act applies, the **offence number**. One scenario involves **no offence at all** — spot it and explain why.

#	Scenario	Full Act (+ year)	CMA offence (1/2/3) or "n/a"
1	An employee guesses a colleague's password and reads their private emails "out of curiosity", changing nothing.		
2	A teenager writes and releases a virus that deletes files on infected machines.		
3	A penetration tester, under a signed contract with the company, breaks into its servers to find weaknesses.		
4	An attacker breaks into an online shop's database to copy card details, planning to sell them.		
5	An advertising agency uses a photographer's image from her website in a national campaign without permission or payment.		
6	A quiz app secretly uploads users' entire contact lists to a data broker without telling them.		
7	A government agency intercepts a suspect's phone calls after obtaining the legally required authorisation . Which Act <i>makes this lawful</i> ?		

Which scenario involves **no offence**, and what single word explains why? _____

Challenge box

A hacker breaks into an online retailer's servers and steals a database of customer card details. The retailer had stored that database **unencrypted** and had kept it for years longer than needed.

(a) Which **two** Acts are relevant here, and **why each**?

(b) Which CMA offence(s) has the hacker committed? Explain.

Section B — Exam practice (30 marks)

Answer **all** questions. Questions marked with an asterisk (*) are marked for the quality of your extended response. Where a question refers to the scenario, your answer **must** be applied to FitPulse — generic answers gain few marks.

Scenario — FitPulse. FitPulse is a UK company that sells fitness wristbands with a companion smartphone app. The app continuously records each user's **heart rate, sleep patterns and GPS location**, and stores this on FitPulse's servers. FitPulse is planning a partnership in which users can share their fitness data with an **insurance company** in exchange for lower premiums; users who refuse will pay the standard (higher) price. Recently: a **hacker** gained access to FitPulse's customer database and viewed records but changed nothing; and a **former employee** has kept a copy of FitPulse's app source code, which she intends to use to launch a rival app.

Q1. State **two** principles of the Data Protection Act 2018 that FitPulse must follow when storing its users' data. **[2]** (AO1)

Q2. The hacker gained access to FitPulse's customer database and viewed records but changed nothing. State the name of the Act the hacker has broken and identify **which offence** under that Act has been committed. **[2]** (AO2)

Q3. Explain why the former employee's plan to reuse FitPulse's source code in a rival app is likely to be unlawful, naming the relevant legislation. **[3]** (AO2)

Q4. FitPulse's app records users' **GPS location** at all times, and the company is considering selling "anonymised" location histories to advertisers. Explain **two** distinct concerns this raises for FitPulse's users. **[4]** (AO2)

Q5. The police ask FitPulse to hand over one user's GPS history as part of a criminal investigation. Describe how the **Regulation of Investigatory Powers Act 2000** governs whether and how this data can be obtained. **[3]** (AO2)

Q6. Some FitPulse customers are elderly or on low incomes and do not own a smartphone capable of running the app, so they cannot access the discounted insurance. Explain how this situation illustrates the **digital divide**, and describe **one** action FitPulse or the insurer could take to reduce its effect. **[4]** (AO2)

Q7* "Schemes in which wearable devices share users' health data with insurance companies do more good than harm." Discuss this statement with reference to FitPulse's planned insurance partnership. In your answer you should consider the benefits and drawbacks for a **range of stakeholders**, and come to a justified conclusion. **[12]** (AO3)

(Total for Section B: 30 marks)
